

Designing Synthetic Training Layouts from Real Hand–Product Priors for Industrial Assembly Vision

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Abstract

Many synthetic-data pipelines rely on random sampling that fails to reproduce the structural distribution of target scenes. We formulate synthetic training layout design as a distribution-alignment problem, shifting the focus from random sampling to structural alignment, and evaluate this perspective in an industrial assembly setting. From real manufacturing annotation, we extract process-specific hand–product priors and use them to generate synthetic frame-level layouts. The full generator combines interaction-aware statistics—entry direction, nearest hand–product distance, overlap, and relative offset—with workspace-aware spatial support. Compared with a count-matched random baseline, it substantially reduces the measured mismatch in key interaction statistics and better aligns hand and product center heatmaps with the real process. These results suggest that prior-guided generation can produce layouts that are more structurally aligned with the target process under a fixed generation budget.

Keywords: synthetic data generation, distribution alignment, layout generation, industrial assembly

1. Introduction

Synthetic data generation is widely used to reduce the cost of collecting and annotating training data for vision systems. However, many pipelines still rely on random placement or broad domain randomization, which can increase appearance diversity while failing to reproduce the structural distribution of real scenes. When the target task depends on how entities are arranged and interact, random sampling can spend a large fraction of a fixed generation budget on layouts that are visually plausible yet structurally irrelevant. Under a fixed budget, this is not only a realism issue but also a practical data-efficiency concern.

Industrial assembly provides a concrete case: a vision system must understand not only whether hand or product appears, but also where products

are concentrated, how hands enter, and how interactions evolve. Random placement diffuses localized work zones and distorts typical entry directions, distances, and overlaps.

These observations motivate a layout-level view of synthetic data generation. Rather than asking only whether a rendered image looks plausible, we ask whether the generated layout distribution matches the target process. We formulate synthetic layout generation as a distribution-alignment problem, introduce a prior-guided generator that models both interaction-level and workspace-level structure, and show in our industrial assembly case that interaction-aware modeling alone is insufficient and that workspace-aware priors provide additional alignment beyond it under a fixed budget.

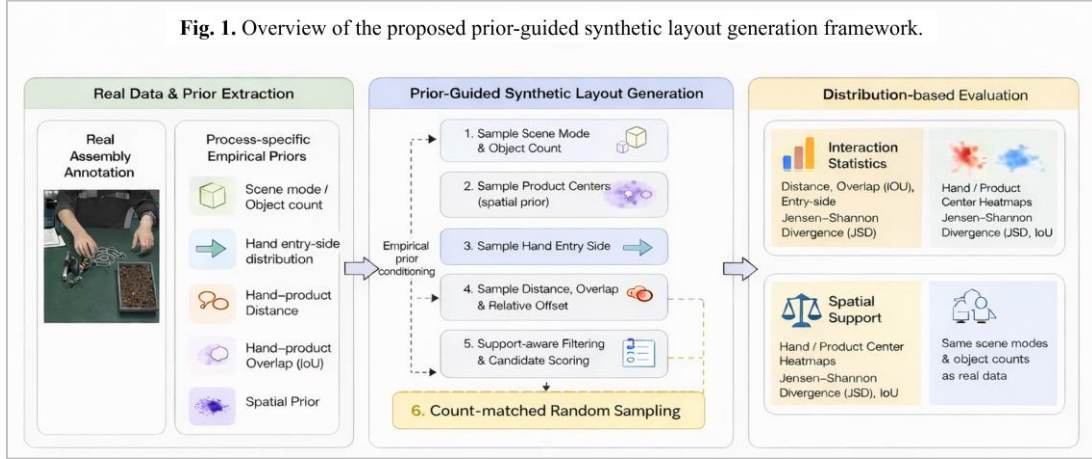


Fig. 1. Overview of the proposed prior-guided synthetic layout generation framework.

2. Related Work

Synthetic data and domain randomization improve robustness by varying appearance factors and enabling scalable annotation pipelines [1], [2], [12]. However, most of this literature targets appearance diversity or sim-to-real robustness rather than whether generated layouts follow the structural distribution of a target process.

In industrial vision, synthetic and limited-data learning have been effective for inspection, classification, and anomaly detection [3]–[6],

[12]. Assembly settings differ because the spatial organization of hand–product interaction within a constrained workspace is itself important.

Assembly datasets and HOI research show that repeated hand–object relations are central to scene understanding [7]–[11]. Prior work mainly uses such structure for recognition or inference. Our contribution lies at the generation stage: we treat synthetic layout generation as a distribution-alignment problem and use process-specific hand–product priors to align both interaction structure and workspace support under a fixed generation budget.

Table 1. Comparison with related work.

Study	Main Focus	Use of Real Interaction Prior	Layout Structure Modeling	Stage
Domain Randomization [1]	Appearance variation for sim-to-real robustness	×	×	Data generation
BlenderProc [2]	Scalable synthetic data pipeline	×	×	Data generation
Industrial Synthetic Data [12]	Inspection / classification in manufacturing	×	Partial	Training
Assembly Interaction Datasets [7]–[9]	Hand–object interaction understanding	✓ (analysis)	✓	Recognition
This paper	Synthetic layout design for assembly vision	✓	✓	Generation

3. Problem Formulation and Method

3.1 Problem formulation

Let H_f and P_f denote the sets of hand and product bounding boxes in frame f . A frame is summarized by a scene descriptor z_f that contains object counts, hand and product center distributions, nearest hand–product distance, maximum hand–product IoU, and hand entry-side.

$$z_f = \phi(H_f, P_f) = \{n_h^f, n_p^f, C_h^f, C_p^f, D_f, U_f, E_f\}$$

Using this descriptor, the real target process induces a process-specific empirical $P_{\text{real}}^{\text{proc}}(z)$, and a synthetic generator G induces its own distribution $P_{\text{syn}}^G(z)$. In this study, the target process is an industrial assembly process, and the objective is to minimize the mismatch between real and synthetic structural distributions at the layout-descriptor level. This formulation emphasizes alignment of scene descriptors rather than pixel-level similarity or random sample volume.

$$G^* = \arg \min_G D(P_{\text{syn}}^G(z), P_{\text{real}}^{\text{proc}}(z))$$

The extracted prior is process-specific, conditioned on the analyzed workcell and task.

3.2 Real prior extraction

We extract empirical priors at three levels: (i) scene-level counts and coarse modes, (ii) interaction-level hand entry-side, nearest hand–product distance, maximum IoU, and relative offset, and (iii) workspace-level product-center and conditional hand-center support.

3.3 Count-matched random baseline

The baseline samples counts from empirical distributions but places objects without interaction or workspace priors, serving as a minimal count-matched control.

3.4 Proposed generators

Proposed (interaction-aware) models scene-mode-conditional counts, hand entry-side, nearest hand–product distance, maximum overlap, and relative offset, reproducing pairwise interaction statistics without explicit spatial support constraints.

The full model, Proposed (interaction + workspace), adds empirical product-center and conditional hand-center priors with workspace support masks, restricting placement to regions consistent with the real workcell.

Each frame is generated by sampling mode and counts, proposing product placement, sampling interaction statistics, and instantiating candidate hand positions. In the full model, candidates are additionally filtered by support masks and scored by spatial priors before acceptance.

4. Experimental Setup

We evaluate on the KAMP industrial assembly dataset: a single production-line workcell captured by a fixed top-down camera at 1,920×1,080 resolution.

The annotated corpus contains 5,020 frames with 68,699 bounding-box annotations (57,564 product, 11,135 hand instances), averaging 10.6 ± 5.5 products and 2.1 ± 1.7 hands per frame; hands are absent in 24.5% of frames.

Frames are tagged across four assembly phases (approach $n=3,903$; contact $n=4,594$; idle $n=212$; release $n=2,247$) with additional hand entry-side labels (left, bottom, top, right). The workspace region occupies $x \in [9.3\%, 80.5\%]$ and $y \in [41.5\%, 90.2\%]$ of the frame.

This dataset provides both the empirical priors and the real reference distribution. Each generator produces 5,000 synthetic frames under a fixed budget.

For quantitative evaluation, we use Jensen–Shannon divergence (JSD) for three primary distributions: nearest hand–product distance, maximum hand–product IoU, and hand entry-side. These metrics capture proximity, contact tendency, and entry behavior. As secondary evidence, we use hand and product center heatmaps to inspect spatial support visually.

5. Results and Discussion

5.1 Main quantitative result

The quantitative comparison in Table 2 confirms that matching object counts alone does not guarantee alignment of structural scene distributions. Relative to Baseline, Proposed reduces distance JSD from 0.279 to 0.0086, IoU JSD from 0.435 to 0.0163, and entry-side JSD from 0.185 to 0.0067. The mean nearest hand–product distance moves from 191.05 pixels to 81.91 pixels, close to the real value of 91.79, while the mean IoU rises from 0.0028 to 0.1163, close to the real value of 0.1077. Thus, the proposed method improves alignment with key

interaction distributions rather than merely matching counts.

Table 2. Main quantitative comparison between Baseline and Proposed (interaction + workspace).

Comparison	Distance JSD	IoU JSD	Entry JSD	Synthetic mean distance	Synthetic mean IoU
Real vs Baseline	0.279	0.435	0.185	191.05	0.0028
Real vs Proposed	0.0086	0.0163	0.0067	81.91	0.1163

Under a fixed synthetic budget, such alignment suggests that a larger share of generated layouts may be consistent with the target process distribution before rendering or downstream learning.

Fig. 2 visualizes hand and product center heatmaps to assess spatial alignment beyond the primary interaction metrics. Baseline spreads

objects broadly, whereas Proposed localizes them toward the real work-zone structure.

The same tendency appears in product placement. Product support strongly affects hand support because hand placement is partly determined by where products are located. The full model therefore improves structural similarity by constraining both what interactions occur and where they are likely to occur in the workspace.

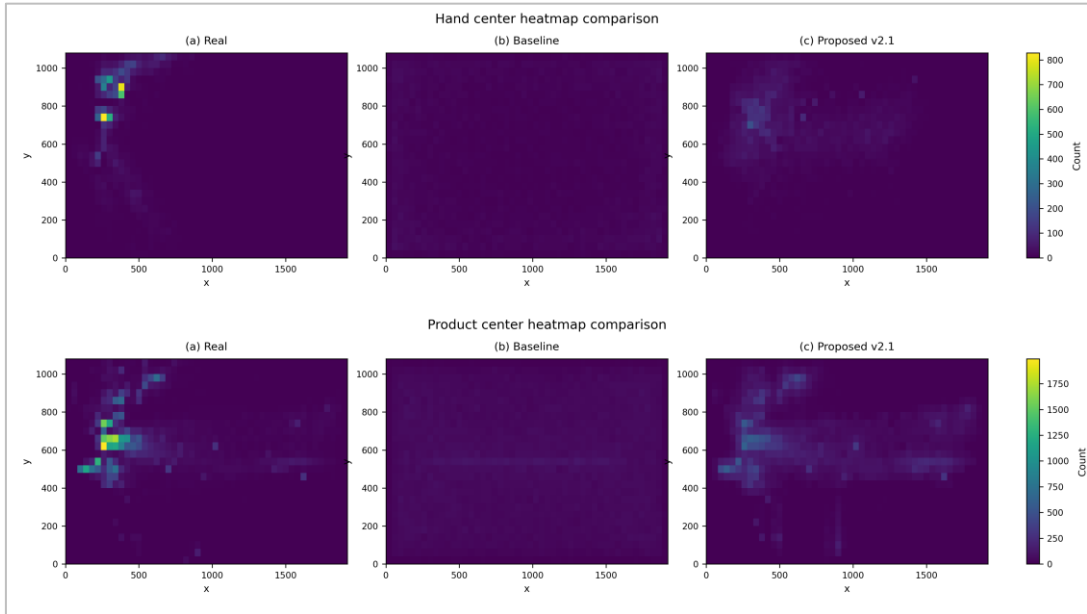


Fig. 2. Spatial distribution comparison of hand and product centers.

5.2 Ablation study

Table 3 ablates the contribution of each prior component starting from a random baseline (distance JSD = 0.058, entry JSD = 0.133).

Workspace constraints dominate distance alignment (heuristic_workspace: distance JSD = 0.018), while entry-side priors dominate entry alignment (prior_entry_only: entry JSD =

8.7×10^{-5}), confirming that each prior component targets a distinct structural property.

Combining priors achieves the best overall alignment: prior_full attains entry JSD = 2.3×10^{-5} with IoU JSD = 0.026, validating that decomposing the target distribution into interpretable components leads to complementary structural gains.

5.3 Generalizability, limitations, and discussion

These results should also be interpreted carefully. Although Proposed improves structural alignment relative to Baseline, the synthetic layouts are not identical to the real distribution, the hand heatmap still has broader support than the real one, and the evaluation is conducted at Downstream task evaluation. The KAMP dataset provides only JSON-format annotations; the corresponding image data (~320 GB) is held by the partner company and was not made available for external use, making end-to-end detector evaluation infeasible. Evaluation is therefore conducted at the distribution level via JSD and Wasserstein metrics. Downstream validation once image data becomes accessible through institutional agreement remains a priority for future work.

Generalizability. The study is conducted on a single workcell; extracted priors are process-specific and not assumed to transfer universally. The framework is formulated to generalize — any annotated process can supply the same prior components — but whether structural patterns transfer across workcells, camera configurations, or assembly procedures remains an open question. Processes with constrained, repetitive interactions are expected to be most amenable; multi-workcell and cross-domain transfer is a clear direction for future work.

the layout level without a downstream task benchmark. The extracted priors are process-specific rather than universal, and the primary evaluation metrics partially overlap with the statistics used to guide generation; future work should therefore include stronger baselines, additional independent descriptors, and downstream validation.

Novelty relative to structural generation approaches. Unlike end-to-end learned generative models (GANs, VAEs, diffusion-based layouts), this work explicitly decomposes the scene distribution into interpretable components — count, interaction, and workspace — composed through conditional sampling. This trades expressive capacity for interpretability and data efficiency: each prior is estimated from modest annotation without deep model training. The primary contribution is demonstrating that explicit prior decomposition achieves strong structural alignment in the industrial assembly setting.

Two independent measures confirm distribution alignment beyond the primary JSD priors. Product-center heatmap JSD decreases from 0.299 (baseline) to 0.174, with L1 heatmap error reducing from 1.270 to 0.884 (42% and 30% improvements). Wasserstein proximity distance reduces from 99.3 px to 26.5 px (73%), corroborating alignment through an independent transport-cost measure not used in prior design.

Table 3. Ablation analysis of count, interaction, and workspace priors.

Method	Count prior	Interaction prior	Workspace prior	Interaction alignment	Spatial support
Baseline	marginal count matching	none	none	Low	diffuse support
Proposed (interaction-aware)	scene-mode conditional count	entry-side, distance, IoU, relative offset	none	High	broad support
Proposed (interaction + workspace)	scene-mode conditional count	same interaction prior as above	product-center heatmap, hand-center conditional heatmap, support mask, candidate scoring	High (preserved)	localized to real hotspots

6. Conclusion

This paper formulated synthetic layout generation as a distribution-alignment problem. By extracting process-specific hand–product priors, we constructed a generator targeting structural scene distributions. In our setting, count-matched random generation was insufficient to reproduce the observed structure of the real assembly scenes.

Proposed (interaction + workspace) outperformed the interaction-only variant by additionally aligning the spatial support of interactions, confirming that interaction-aware and workspace-aware priors play complementary roles in layout-level distribution alignment.

Future work will connect the proposed layout priors to downstream model training, stronger baselines, richer descriptors, and transfer across workcells, camera settings, and assembly procedures.

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